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Splunk SSH Authentication Monitoring & Log Analysis

1. Objective

The objective of this lab was to deploy Splunk Enterprise, ingest Linux authentication logs, and investigate SSH-related security events. The analysis focused on monitoring successful logins, failed authentication attempts, privileged command execution, and overall SSH activity through centralized log collection.

Summary

A centralized log monitoring lab was created using Splunk Enterprise running on Ubuntu. Authentication logs from `/var/log/auth.log` were continuously ingested into Splunk and analyzed for SSH-related activity.

The investigation identified both failed and successful SSH authentication events originating from the Kali Linux host (192.168.56.10) targeting the Ubuntu server (192.168.56.11). Multiple failed password attempts were recorded before a successful login was achieved. Privileged activities executed through `sudo` were also captured and analyzed.

The collected evidence demonstrates how authentication logs can be used to reconstruct user activity, identify brute-force behavior, validate successful access, and monitor administrative actions.

2. Lab Environment

Component	Details
SIEM Platform	Splunk Enterprise
Target Host	Ubuntu
Log Source	Ubuntu Linux
Client System	Kali Linux
Service Monitored	OpenSSH
Log File	<code>/var/log/auth.log</code>
Network Type	VirtualBox Internal Network

3. Splunk Installation & Configuration

Splunk Enterprise was installed on the Ubuntu server and configured for local access through the web interface. After installation, the Splunk service was started and verified to ensure successful operation.

```

vboxuser@ubuntu:~$ sudo dpkg -i splunk-10.4.0-f798d4d49089-linux-amd64.deb
[sudo: authenticate] Password:
(Reading database. 140495 files and directories currently installed.)
Preparing to unpack splunk-10.4.0-f798d4d49089-linux-amd64.deb.
verify that this system has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunk (10.4.0) over (10.4.0).
Setting up splunk (10.4.0).
find: '/opt/splunk/lib/python3.7/site-packages': No such file or directory
complete
vboxuser@ubuntu:~$ sudo /opt/splunk/bin/splunk start
Running Splunk Enterprise as root is deprecated and will be removed in a future release. For details, see the Release Notes. To run as root, use the --run-as-root option.
vboxuser@ubuntu:~$ _

vboxuser@ubuntu:~$ sudo /opt/splunk/bin/splunk start --run-as-root

Splunk> Winning the War on Error

Checking prerequisites...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8065]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
  Checking critical directories... Done
  Checking indexes...
    Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _thefishbucket history main summary
    Done
  Checking filesystem compatibility... Done
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunk/splunk-10.4.0-f798d4d49089-linux-amd64-manifest'
  All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

Waiting for web server at http://127.0.0.1:8000 to be available..... Done

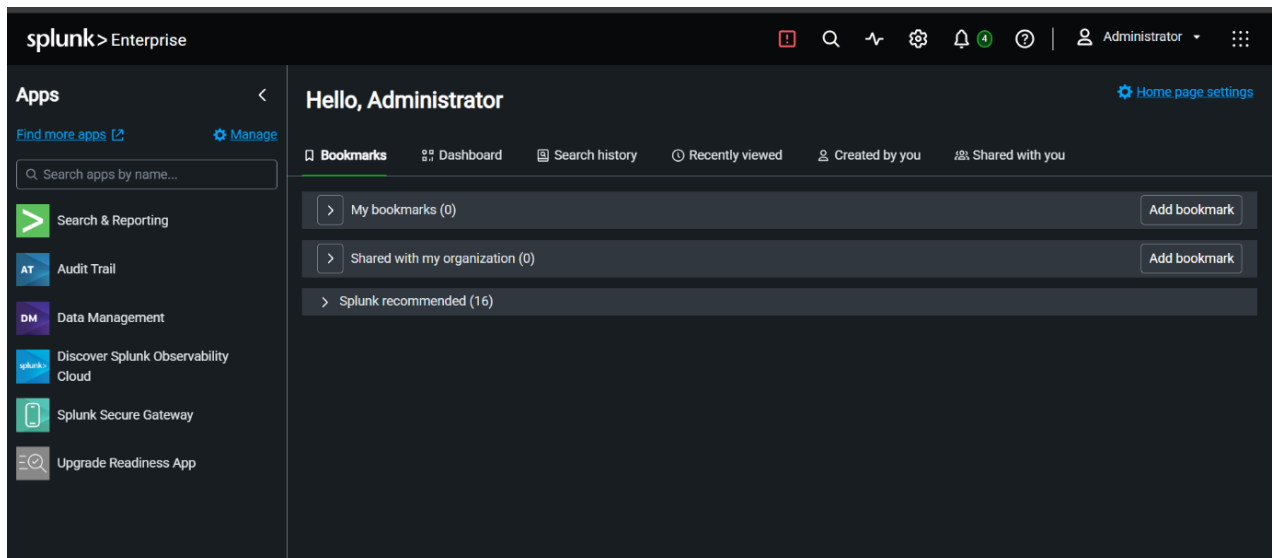
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://ubuntu:8000

```

Observation

The Splunk web dashboard became accessible and ready to receive security log data for analysis.



4. Data Input Configuration

A File Monitor data input was configured to continuously monitor:

/var/log/auth.log

The monitored log source was assigned to the Linux authentication log category, allowing authentication events to be indexed automatically.

The screenshot shows the 'Files & Directories' configuration page in Splunk. On the left, there are three tabs: 'Files & Directories' (selected), 'HTTP Event Collector', and 'TCP / UDP'. The 'Files & Directories' tab has a description: 'Upload a file, index a local file, or monitor an entire directory.' The 'HTTP Event Collector' tab has a description: 'Configure tokens that clients can use to send data over HTTP or HTTPS.' The 'TCP / UDP' tab has a description: 'Configure the Splunk platform to listen on a network port.' The main content area on the right has a heading 'Configure this instance to monitor files and directories for data. To monitor all objects in a directory, select the directory. The Splunk platform monitors and assigns a single source type to all objects within the directory. This might cause problems if there are different object types or data sources in the directory. To assign multiple source types to objects in the same directory, configure individual data inputs for those objects. [Learn More](#)'. Below this, there is a 'File or Directory?' label, a text input field containing '/var/log/auth.log', and a 'Browse' button. At the bottom, there is a note: 'On Windows: c:\apache\apache.error.log or \\hostname\apache\apache.error.log. On Unix: /var/log or /mnt/www01/var/log.'

The screenshot shows the 'Review' section of the Splunk configuration. It lists the following configuration details:

Input Type	File Monitor
Source Path	/var/log/auth.log
Continuously Monitor	Yes
Source Type	linux_messages_syslog
App Context	search
Host	ubuntu
Index	default

Observation

Authentication logs generated on the Ubuntu system were successfully ingested into Splunk and became searchable through SPL queries.

5. Log Ingestion Verification

After configuring the file monitor, searches were performed to verify that authentication events were being indexed correctly.

To establish a baseline and confirm data pipeline integrity, a broad historical query was executed across the environment.

```
index=*
```

The search interface successfully returned a total of 214 events.

The screenshot shows the Splunk search interface for the query `index=*`. The search results show 214 events. The interface includes a search bar, a timeline visualization, and a list of events. The events are displayed in a table with columns for Time and Event. The events are sorted by Time, showing a range from 6/21/26 12:17:02.906 PM to 6/21/26 11:43:53.972 AM. The events include details about user sessions, such as session closure and opening for user root.

Time	Event
6/21/26 12:17:02.906 PM	2026-06-21T12:17:02.906044+00:00 ubuntu CRON[68469]: pam_unix(cron:session): session closed for user root
6/21/26 12:17:02.520 PM	2026-06-21T12:17:02.520091+00:00 ubuntu CRON[68469]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
6/21/26 11:43:53.972 AM	2026-06-21T11:43:53.972484+00:00 ubuntu systemd-logind[64894]: New session '2' of user 'vboxuser' with class 'manager' and type 'unspecified'.
6/21/26 11:43:53.972 AM	2026-06-21T11:43:53.972417+00:00 ubuntu systemd-logind[64894]: New session '1' of user 'vboxuser' with class 'user' and type 'tty'.

To verify the actions and session durations of connections, the search scope was expanded generally to include all core SSH daemon processes.

```
index=* sshd
```

The screenshot shows the Splunk search interface for the query `index=* sshd`. The search results show 28 events. The interface includes a search bar, a timeline visualization, and a list of events. The events are displayed in a table with columns for Time and Event. The events are sorted by Time, showing a range from 2026-06-20T17:54:18.608992+00:00 to 2026-06-20T17:54:12.359360+00:00. The events include details about SSH sessions, such as session closure, disconnection, and login/logout events.

Time	Event
2026-06-20T17:54:18.608992+00:00	ubuntu sshd-session[2312]: pam_unix(sshd:session): session closed for user vboxuser
2026-06-20T17:54:18.607372+00:00	ubuntu sshd-session[2369]: Disconnected from user vboxuser 192.168.56.10 port 46990
2026-06-20T17:54:18.603057+00:00	ubuntu sshd-session[2369]: Received disconnect from 192.168.56.10 port 46990:11: disconnected by user
2026-06-20T17:54:18.597488+00:00	ubuntu sshd-session[2312]: syslogin_perform_logout: logout() returned an error
2026-06-20T17:54:12.359360+00:00	ubuntu sshd-session[2312]: pam_unix(sshd:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)

Observation

Multiple SSH and system authentication events appeared within the Splunk search interface, confirming successful log collection.

Security Relevance

Verifying log ingestion is essential because missing or incomplete logs can prevent analysts from detecting unauthorized access attempts.

6. Incident Analysis & Event Tracking

6.1 Raw Inspection of Failed Authentication Attempts

To investigate potential unauthorized access or brute-force tracking, the index was queried for failed password strings.

index=* "Failed password"

The screenshot shows the Splunk search interface. The search bar contains the query `index=* "Failed password"`. The results show 8 events. The left sidebar displays the search configuration and field lists. The main panel shows a list of events with details such as timestamp, host, source, and the failed password attempt.

Time	Host	Source	Event
2026-06-20T17:28:36.610671+00:00	ubuntu	sudo: vboxuser	TTY=/dev/tty1 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-06-19T18:32:17.432875+00:00	ubuntu	sshd-session[4045]	Failed password for vboxuser from 192.168.56.10 port 49780 ssh2
2026-06-19T18:32:08.511806+00:00	ubuntu	sshd-session[4040]	Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:32:05.144256+00:00	ubuntu	sshd-session[4040]	Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:32:00.589768+00:00	ubuntu	sshd-session[4040]	Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19T18:31:52.830122+00:00	ubuntu	sshd-session[4035]	Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19T18:31:48.200108+00:00	ubuntu	sshd-session[4035]	Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19T18:31:43.161195+00:00	ubuntu	sshd-session[4035]	Failed password for vboxuser from 192.168.56.10 port 57198 ssh2

Observation

The raw log analysis explicitly captures a series of authentications rejected by the system daemon (sshd-session). The logs detail the source port variations typical of rapid, automated login attempts originating from a single host machine (192.168.56.10).

- **Total Captured Anomalies:** 8 events
- **Target Account:** vboxuser
- **Attacking Source IP:** 192.168.56.10

Analysis

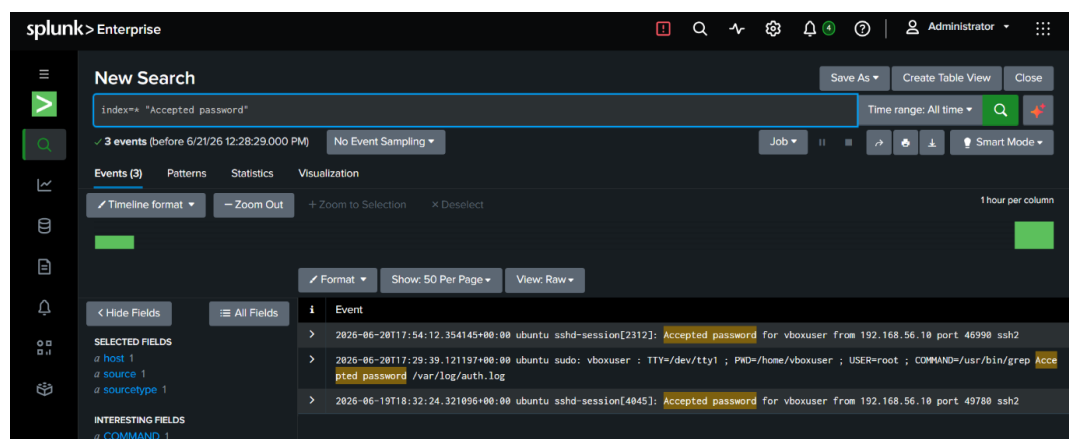
Failed password events indicate unsuccessful authentication attempts. These events are commonly reviewed during investigations involving:

- **Incorrect credentials**
- **Password guessing**
- **Brute-force attempts**
- **Unauthorized access attempts**

6.2 Raw Inspection of Successful Authentications

A critical phase of incident scoping involves identifying whether any malicious attempts successfully bypassed system security. The index was queried for successful login indicators.

index=* "Accepted password"



Observation

The search returned three distinct logs confirming accepted credentials. Of immediate concern to an analyst is the log matching timestamp 2026-06-19 18:32:24, where the password was accepted for vboxuser from the same attacking IP address (192.168.56.10) exactly **7 seconds after the final recorded failure** in the brute-force sequence.

- **Total Successful Sessions:** 3 events
- **Target Account:** vboxuser
- **Authorized IP Address:** 192.168.56.10

Analysis

Successful authentication events establish a timeline of legitimate access and help validate user activity during investigations.

6.3 Tabular Analysis and Chronological Sorting of Failures and Successful sessions

To reconstruct the exact timeline of the unauthorized login attempts and timeline of successful access events, the unstructured log data was formatted into a clean, chronologically sorted data table.

`index=* "Failed Password" | table _time _raw | sort - _time`

The screenshot shows a Splunk search interface with the query `index=* "Failed Password" | table _time _raw | sort - _time`. It displays 8 events from before 6/21/26 2:59:22.000 PM. The results table has two columns: `_time` and `_raw`. The data shows multiple failed password attempts for the `vboxuser` from IP `192.168.56.10` on port `49760` using `ssh2`. The log messages are truncated in the `_raw` column.

_time	_raw
2026-06-20 17:28:35.618	2026-06-20T17:28:35.618Z+00:00 ubuntu sudo: vboxuser : TTY=/dev/ttyl ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-06-19 18:32:17.432	2026-06-19T18:32:17.432Z+00:00 ubuntu sshd-session[4045]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19 18:32:08.511	2026-06-19T18:32:08.511Z+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19 18:32:05.144	2026-06-19T18:32:05.144Z+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19 18:32:00.589	2026-06-19T18:32:00.589Z+00:00 ubuntu sshd-session[4040]: Failed password for vboxuser from 192.168.56.10 port 49760 ssh2
2026-06-19 18:31:52.830	2026-06-19T18:31:52.830Z+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19 18:31:48.200	2026-06-19T18:31:48.200Z+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2
2026-06-19 18:31:43.161	2026-06-19T18:31:43.161Z+00:00 ubuntu sshd-session[4035]: Failed password for vboxuser from 192.168.56.10 port 57198 ssh2

This structural query strips away non-essential interface fields and presents the raw payload alongside a unified timestamp (`_time`). The resulting analysis shows a dense cluster of seven authentication failures occurring within a 34-second window.

`index=* "Accepted Password" | table _time _raw | sort - _time`

The screenshot shows a Splunk search interface with the query `index=* "Accepted Password" | table _time _raw | sort - _time`. It displays 3 events from before 6/21/26 3:00:25.000 PM. The results table has two columns: `_time` and `_raw`. The data shows successful login attempts for the `vboxuser` from IP `192.168.56.10` on port `46990` using `ssh2`. The log messages are truncated in the `_raw` column.

_time	_raw
2026-06-20 17:54:12.354	2026-06-20T17:54:12.354Z+00:00 ubuntu sshd-session[2312]: Accepted password for vboxuser from 192.168.56.10 port 46990 ssh2
2026-06-20 17:29:39.121	2026-06-20T17:29:39.121Z+00:00 ubuntu sudo: vboxuser : TTY=/dev/ttyl ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/grep Accepted password /var/log/auth.log
2026-06-19 18:32:24.321	2026-06-19T18:32:24.321Z+00:00 ubuntu sshd-session[4045]: Accepted password for vboxuser from 192.168.56.10 port 49760 ssh2

Tabular Auditing of Successful Sessions

The structured table provides an irrefutable trail of system access points:

1. **First Access:** June 19, 2026, at 18:32:24 (Directly following the brute-force attempt).
2. **Second Access:** June 20, 2026, at 17:29:39 (An administrative session via a `sudo grep` command).
3. **Third Access:** June 20, 2026, at 17:54:12 (A new, separate `sshd-session` established on port 46990).

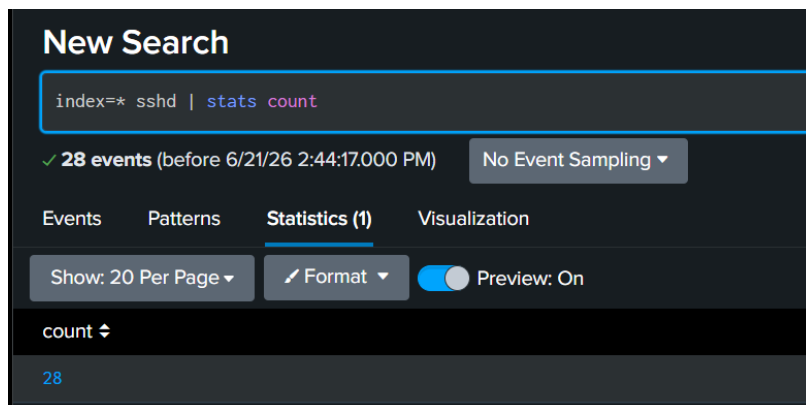
7. Data Aggregation, Field Profiling & Trends

This section focuses on compiling raw events into higher-level numbers, profiling metadata fields, and bucketizing time intervals to establish actionable intelligence.

7.1 Total Metric Volumetrics

To quickly evaluate overall exposure during high-level reporting cycles, a basic aggregation was applied to the SSH processes.

`index=* sshd | stats count`

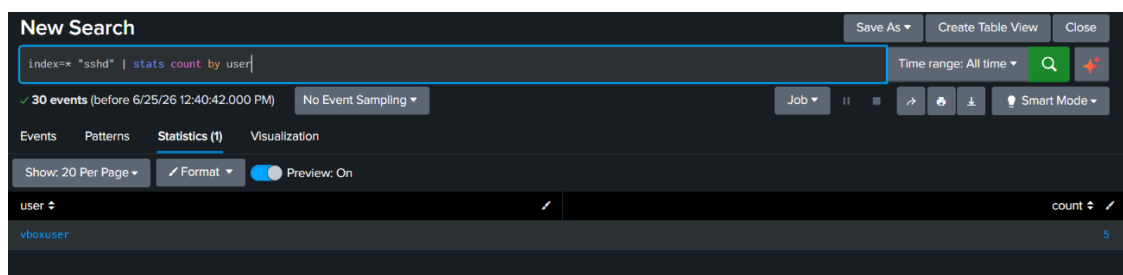


The query aggregates all matching instances, yielding a total of **28 unique SSH events** recorded across the global log collection timeframe.

7.2 User Volume Distribution

To determine which accounts were the primary targets of authentication traffic within the system daemon logs, a conditional statistical aggregation was applied.

`index=* "sshd" | stats count by user`

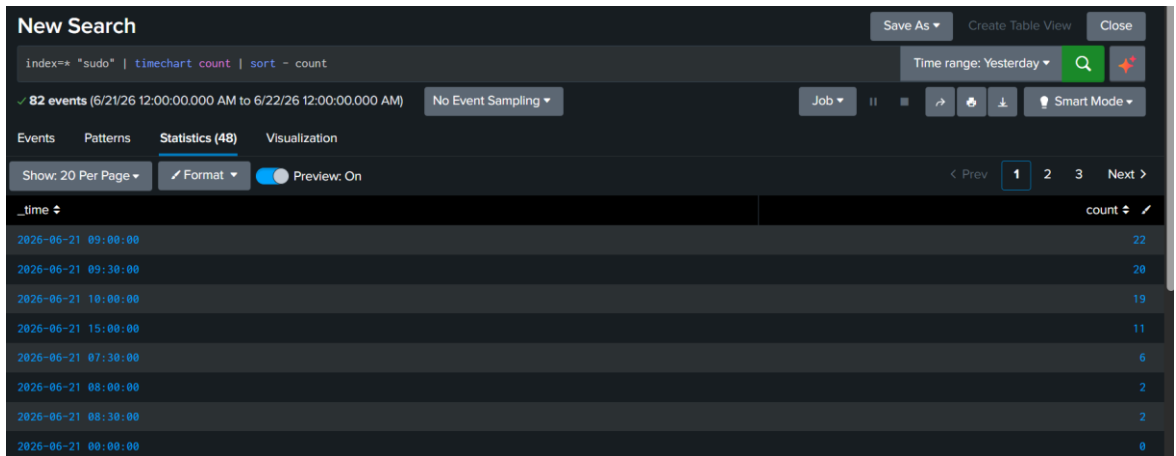


The resulting data panel collapses unstructured data into a user distribution vector, mapping exactly **5 distinct signature-matched events** cleanly to the target account vboxuser. This metric allows incident handlers to confirm that password guessing was not scattered blindly across multiple local system accounts, but was instead explicitly focused on this specific user identity.

7.3 Privileged Escalation Chronological Auditing

To audit administrative behaviors and track command velocity across the server architecture, commands executed with elevated privileges (sudo) were analyzed using time-bucketed aggregations sorted by peak operational intervals.

index=* "sudo" | timechart count | sort - count



- **Total Captured Privilege Events:** 82 events
- **Peak Activity Windows:** June 21, 2026, at 09:00:00 (22 events) and 09:30:00 (20 events)

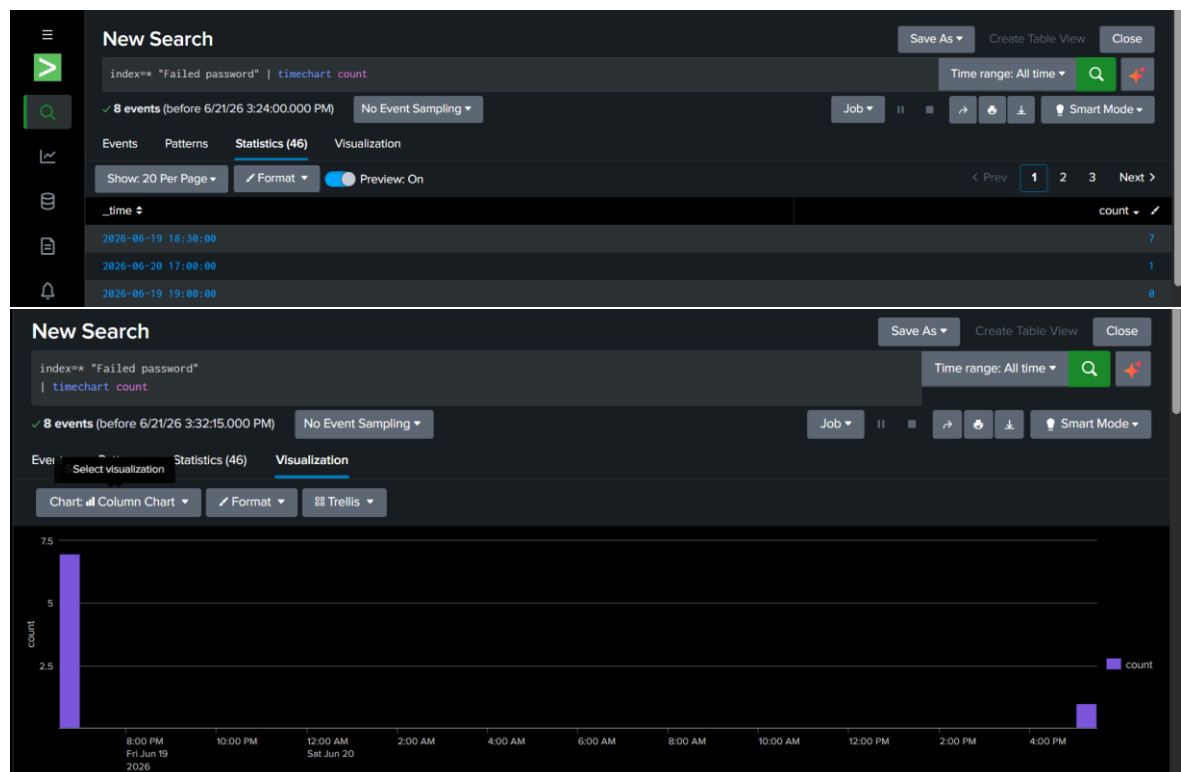
Aggregating sudo activity into density buckets allows security operations teams to establish an operational baseline, ensuring that rapid successions of privileged to catch potential insider threats or compromised local accounts.

8. Visual Dashboards & Anomalous Trend Reporting

8.1 Executive Visualization of Brute-Force Activity

To make the statistical findings easily digestible for Security Operations Center (SOC) dashboards and executive reviews, the data chart was converted into a visual representation.

`index=* "Failed password" | timechart count`



Observation

Analytical Insights from Visualization

- **Anomalous Spike Isolation:** The column chart provides immediate visual confirmation of an aggressive authentication anomaly on **Friday, June 19, 2026**, where failed attempts sharply spike to their peak value of 7.
- **Incident Response Value:** Utilizing these automated Splunk visualizations enables SOC teams to establish clear behavioral baselines and immediately identify dangerous spikes in authentication traffic that require active security intervention.

Security Relevance

Sudo events provide visibility into privileged account usage and administrative activity. Monitoring privileged actions helps identify misuse of elevated permissions and supports incident investigations.

9. Findings

The system investigation produced the following key findings:

1. **Successful Log Ingestion:** Authentication logs were dynamically and successfully ingested into Splunk.
2. **Threat Visibility:** Failed SSH login attempts were efficiently identified, isolated, and investigated.
3. **Access Auditability:** Successful SSH authentication events were tracked to build clear timelines of user access.
4. **Targeted Vector Identification:** Isolated telemetry confirmed that password guessing campaigns focused directly on the vboxuser account rather than general credential sprays.
5. **Privileged Escalation Monitoring:** Critical administrative sudo activity was audited through centralized log tracking, revealing key operational usage bursts.
6. **Operational Intelligence:** Statistical searches and visualizations significantly enhanced event trend analysis.
7. **Centralized Security Posture:** Transitioning to centralized logging drastically improved overall visibility into authentication-related system activities.

10. Conclusion

This lab successfully demonstrated the deployment, configuration, and practical application of **Splunk Enterprise** for centralized SSH authentication monitoring. By collecting, indexing, and analyzing authentication logs from an Ubuntu server, the deployment successfully uncovered failed logins, validated authorized access, and exposed privileged command activities.

Ultimately, this exercise highlighted the fundamental value of **SIEM platforms** in automating log aggregation, streamlining security monitoring, and accelerating incident response workflows.